

Nate Thach (Nguyen Thach)

✉ nate.thach@huskers.unl.edu

📍 University of Nebraska-Lincoln (UNL), Lincoln, NE, USA

🔗 Google Scholar

🆔 0009-0003-3308-524X

🌐 LinkedIn

Research Interest

Novel Applications of Large Language Models in Computational Social Choice.

Participatory Budgeting, Facility Location, Urban Planning, and Mechanism Design Without Money.

Education

2022 – 2027	📖 Ph.D. in Computer Science Advisor: Hau Chan	UNL
2017 – 2021	📖 B.S. in Physics and Mathematics Computer Science Minor	UNL

Publications

Conference Proceedings

- 1 **N. Thach**, H. Chan, D. Parkes, and K. Lakhani, “Guiding worker self-selection in crowdsourcing contests: An llm-augmented algorithmic approach,” in *The 2026 ACM Conference on Human-AI Complementarity and Alignment (HCOMP 2026)*, forthcoming, 2026.
- 2 **N. Thach**, F. Liu, H. Zhou, and H. Chan, “Multi-facility location mechanism design with large language models,” in *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*, forthcoming, 2026.
- 3 **N. Thach**, X. Sha, and H. Chan, “Large language models for designing participatory budgeting rules,” in *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems*, ser. AAMAS ’26, IFAAMAS, 2026, pp. 174–182, ISBN: 9798400723179. 🔗 DOI: 10.65109/LGEP3560.
- 4 **N. Thach**, P. Habecker, A. Eisenbraun, W. A. Mason, K. Tyler, B. Khan, and H. Chan, “Muhboost: Multi-label boosting for practical longitudinal human behavior modeling,” in *International Conference on Representation Learning (ICLR)*, 2025, pp. 81 215–81 243.
- 5 **N. Thach**, P. Habecker, B. Johnston, L. Cervantes, A. Eisenbraun, A. Mason, K. Tyler, B. Khan, and H. Chan, “A novel gan approach to augment limited tabular data for short-term substance use prediction,” in *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI-24, AI for Good, IJCAI*, Aug. 2024, pp. 7474–7482.

Journal Articles

- 1 **N. Thach**, C. Wang, and H. Chan, “Improving the location of facilities through network addition modification: A revisit,” *Annals of Operations Research*, vol. 356, no. 2, pp. 1231–1280, Jan. 2026, ISSN: 1572-9338. 🔗 DOI: 10.1007/s10479-025-06869-1.
- 2 **N. Thach**, P. Habecker, B. Johnston, L. Cervantes, A. Eisenbraun, A. Mason, K. Tyler, B. Khan, and H. Chan, “Analyzing and predicting short-term substance use behaviors of persons who use drugs in the great plains of the u.s.,” *PLOS ONE*, vol. 19, no. 11, pp. 1–33, Nov. 2024. 🔗 DOI: 10.1371/journal.pone.0312046.
- 3 K. Preuss, **N. Thach**, X. Liang, M. Baine, J. Chen, C. Zhang, H. Du, H. Yu, C. Lin, M. A. Hollingsworth, and D. Zheng, “Using quantitative imaging for personalized medicine in pancreatic cancer: A review of radiomics and deep learning applications,” *Cancers*, vol. 14, no. 7, p. 1654, 2022.

Ongoing Works

<i>Toward End-to-End LLM-Based Automatic Heuristic Design using Reductions</i> N. Thach, N. Huynh, H. Chan	WSDM 2027
<i>Large Language Models for Designing Voting Rules under Approximate Metric Distortion</i> N. Thach and H. Chan	AAMAS 2027
<i>Fairness of Large Language Models in Participatory Budgeting</i> J. Chen, N. Thach, H. Chan	AAMAS 2027
<i>Measuring Proportional Sample Calibration in Pluralistic Language Model Evaluation</i> X. Sha, N. Thach, H. Chan	

Teaching

📖 Introduction to Computational Game Theory, Fall 2026 (with Dr. Hau Chan)	UNL
📖 Data Structures and Algorithms, Fall 2025 (Guest Lecturer for Dr. Hau Chan)	UNL
📖 Elementary General Physics I Lab, Fall 2020	UNL

Academic Experience

Proposal Contributor

NIH R01	📖 Enhancing Substance Use Risk Prediction Using Generative AI-Derived Synthetic Data. \$1,927,175 (Submitted)
NSF	📖 Collaborative Network for Data-Driven High School Mathematics Pathways and College Mathematics Readiness. \$1,500,000 (Submitted)
	📖 Connecting Local Producers to Institutional Buyers: A Digital Platform for Sustainable Food Systems. \$1,500,000 (Declined)

Conference/Workshop

Organizer	📖 OptLearnMAS-26 at AAMAS 2026 (Co-Chair)
Reviewer	📖 AAAI-26/27, IJCAI-24/25/26, NeurIPS-24/26, ICLR-25/26
Presenter	📖 IJCAI-24, AAMAS-26
Attendant	📖 AAAI-23/25, IJCAI-24, ICLR-25

Invited Talks and Presentation

LLMRule: Designing Participatory Budgeting Rules with Large Language Models
Northwestern University, The Theory Seminar, 2026

Large Language Models for Combinatorial Optimization Problems in Transportation
UNL, Transportation Engineering Seminar, 2025

Evaluating the Effectiveness of Machine-Learning Models for Predicting Short-term Substance Use Behaviors
UNL, The Annual Symposium on Substance Use Research (SOSUR), 2024

Awards

📖 HCOMP-26 NSF Student Travel Grant. \$1,500	HCOMP & CI
📖 IJCAI-26 SIGAI Grant. \$1,000	IJCAI
📖 IJCAI-26 Gold Tier Reviewer Distinction. \$1,030	IJCAI
📖 ICLR-25 Financial Assistance. \$425	ICLR
📖 IJCAI-24 Travel Grant. \$555	IJCAI

Awards (continued)

College of Engineering Professional Development Fellowship. \$500	UNL
Global Laureate Scholarship. \$60,000	UNL

Student Mentoring

- Jingwen Chen, PhD Student at UNL, 2026–Present
- Mark Phan, PhD Student at UNL, 2026–Present
- Xingchen Sha, PhD Student at Northwestern University, 2025–Present
- Aida Riahifar, PhD Student at UNL, 2025–2026